



(Left to right) Bhagyesh Tajne, CTO; Rajesh Thakre, MD; Hitendra Valhe, CEO; and Nilesh Wagh, CMO with products

Arduino IDE and ESP-IDF coding. We included STM32 as an entry-level industrial controller, so when students list experience with Arduino, ESP, STM, and Raspberry Pi on their résumé, it stands out. Plus, we offer a full open source learning management system with videos, lectures, and courses. Basic and intermediate courses are free,

while advanced AI-focused videos are paid and created in-house.”

Designing the kit involved many technical hurdles. After five iterations, solving impedance, memory, and buffer issues through forums, mentors, and trial and error, the final kit was stabilised and refined through student workshops.

While discussing the integration

of AI and ML, Bhagyesh adds, “We had to create our own mathematical library, building functions that run directly on the microcontroller. The first challenge was essentially recreating NumPy in a much simpler form, and it is still being improved for wider TinyML use. Apart from library issues, hardware limitations were a major problem. Unlike laptops or computers, microcontrollers have very limited RAM, and sensor data often comes in inconsistent formats, making it difficult to use for model training.”

While discussing revenue, Hitendra says, “The company was registered as a Private Limited in March 2025 and has provided Robotics and AI labs to about 50 schools and colleges since then.”

Though Ardhay emerged as a by-product of their bootstrapped startup, their long-term aim remains building robots for industry, similar to Tesla’s autonomous navigation cars. Currently, they are developing Arya, a receptionist robot with AI. Arya will include machine learning, image and voice recognition, and interactive conversation capability. Orders are already in place, but delivery will follow once development is complete. This reflects their vision of advancing industrial robotics alongside education products.

—Nidhi Agarwal

### 3 “WHY LEAVE YOUR PARKING SPACE IDLE WHEN IT COULD BE EARNING FOR YOU?”

Amid the city’s layered obstacles, two friends set out to rethink urban mobility and discovered that real change began when private property owners unlocked the hidden potential of their empty parking spaces.

“Why leave your parking space idle when it could be earning for you?” This simple question laid the foundation for ParkoBot, a startup addressing one of India’s most persistent urban challenges: parking. In cities where drivers waste fuel

circling for parking spots, while thousands of private spaces sit unused, the founders saw an opportunity to unlock hidden value.

Launched in 2023 by Amrit Choudhury, Raj Kumar Bihani, and Animesh Mukherjee, the ParkoBot founders combine over a decade of

expertise in RFID, IoT, and fintech to create the world’s first personal, IoT-enabled parking barrier system. Their vision is to turn every idle parking slot into a shareable, revenue-generating asset—an ‘Airbnb for parking.’

At its core, ParkoBot is an